

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL A-LEVEL

FURTHER MATHEMATICS

(9665/FM03) Unit FP2 Pure Mathematics

Thursday 30 May 2024

07:00 GMT

Time allowed: 2 hours 30 minutes

Materials

- For this paper you must have the OxfordAQA Booklet of Formulae and Statistical Tables (enclosed).
- You may use a graphical calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- Show all necessary working; otherwise marks may be lost.

For Examiner's Use	
Question	Mark
1	
2	
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TOTAL	



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Answer **all** questions in the spaces provided.

1 The matrix $\mathbf{M}_R = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$ represents the single transformation R

The matrix $\mathbf{M}_S = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ represents the single transformation S

1 (a) (i) Describe fully the single transformation R represented by the matrix $\mathbf{M}_R$

[2 marks]

1 (a) (ii) Describe fully the single transformation S represented by the matrix $\mathbf{M}_S$

[2 marks]

1 (b) The composite transformation T is obtained by carrying out S followed by R

Find the 3×3 matrix $\mathbf{M}_T$ which represents the transformation T

[2 marks]

$\mathbf{M}_T =$ _____

Turn over ►



2 The vectors **a**, **b** and **c** are given by

$$\mathbf{a} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k} \quad \mathbf{b} = 3\mathbf{i} + 3\mathbf{j} + 5\mathbf{k} \quad \mathbf{c} = -3\mathbf{i} + t\mathbf{j} + 9\mathbf{k}$$

where t is a real number.

2 (a) Find $\mathbf{a} \times \mathbf{b}$

[1 mark]

Answer _____

2 (b) In the case when **a**, **b** and **c** are coplanar, use a scalar triple product to find the value of t

[3 marks]

Answer _____

2 (c) In the case when **c** is parallel to $\mathbf{a} \times \mathbf{b}$ find the value of t

[2 marks]

Answer _____



3

$$x^4 - px^2 + qx - r = 0$$

are α , α , α and β

It is given that p , q and r are positive real numbers.

3 (a) Show that $p^2 = 12r$

[4 marks]

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3 (b) Express the product pr in terms of q

[2 marks]

Answer

A curve C has equation $y^2 = 4x$

The arc of C between the points $(3, 2\sqrt{3})$ and $(4, 4)$ is rotated through 2π radians about the x -axis.

Show that the area of the curved surface generated is $\frac{8\pi}{3}(5\sqrt{5}-8)$

[5 marks]

[illegible]

Use an integrating factor to find the general solution of the differential equation

Give your answer in the form $y = f(x)$

[5 marks]

[illegible]
$$y =$$

5

Turn over ►



$$z = \cos \theta + i \sin \theta$$

[3 marks]

[illegible]

[1 mark]

$$z - \frac{1}{z} =$$



6 (b) Hence express $\cos^3 \theta \sin^4 \theta$ in the form

$$p \cos \theta + q \cos 3\theta + r \cos 5\theta + s \cos 7\theta$$

where p, q, r and s are rational numbers.

[5 marks]

[illegible]

$$\cos^3 \theta \sin^4 \theta =$$

9

Turn over ►



7 (a) Show that $\frac{d^2y}{dx^2} = 2\ln(1+x)+3$

$$\frac{d^n y}{dx^n} = (-1)^{n-1} \frac{2(n-3)!}{(1+x)^{n-2}}$$

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- 7 (c) **Hence** find the exact value for the sum of the coefficients of x^{10} and x^{11} in the Maclaurin series expansion for $(1+x)^2 \ln(1+x)$

[2 marks]

Answer _____

Turn over ►



Solve the differential equation

given that $y=1$ when $x=0$ and $\frac{dy}{dx}=5$ when $x=0$

[10 marks]

[illegible]

[illegible]
$$y = \underline{\hspace{10cm}}$$

10

Turn over ►



Find the exact value of the improper integral

showing the limiting process used.

Give your answer in the form $\ln k$ where k is an expression in terms of π

[illegible]

Answer _____

8



10 The plane Π_1 has vector equation $\mathbf{r} \cdot \begin{vmatrix} 1 \\ 3 \end{vmatrix} = 2$

The line L has Cartesian equation $\frac{x-7}{2} = 1-y = \frac{z-3}{4}$

10 (a) Find the acute angle between the line L and the plane Π_1

Give your answer to the nearest 0.1°

[4 marks]

[illegible]

Answer

10 (b) The point P has coordinates $(4, 1, 15)$

Find the shortest distance from P to the line L

Give your answer in an exact form.

[4 marks]



Answer _____

10 (c) The plane Π_2 has Cartesian equation $x - 3y + 2z = 1$

Find a vector equation for the line of intersection of the planes Π_1 and Π_2

Give your answer in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$

[5 marks]

Answer _____



11

$$\mathbf{M} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & c & -11 \\ 2 & -1 & 1 \end{bmatrix}$$

where c is an integer.

11 (a)

Find $\mathbf{M}^{-1}$ in terms of c

[5 marks]

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Answer _____

Question 11 continues on the next page

Turn over ►



where

$$\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3 = -4$$

[5 marks]

[illegible]

[2 marks]

Answer

[3 marks]

[illegible]

Answer

15

Turn over ►



- 12** The five roots of the equation $z^5 = 4 - 4i$ when plotted on an Argand diagram form the vertices of a pentagon.

- 12 (a)** Find the root that is closest to the Imaginary axis.

Give your answer in the form $r e^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$

[3 marks]

Answer _____

- 12 (b)** Find the product of the roots that lie above the Real axis.

Give your answer in the form $r e^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$

[2 marks]

Answer _____



12 (c) Find the perimeter of the pentagon.

Give your answer in the form $a \sin(b\pi)$ where a and b are exact values

[2 marks]

Answer _____

7
--

Turn over for the next question

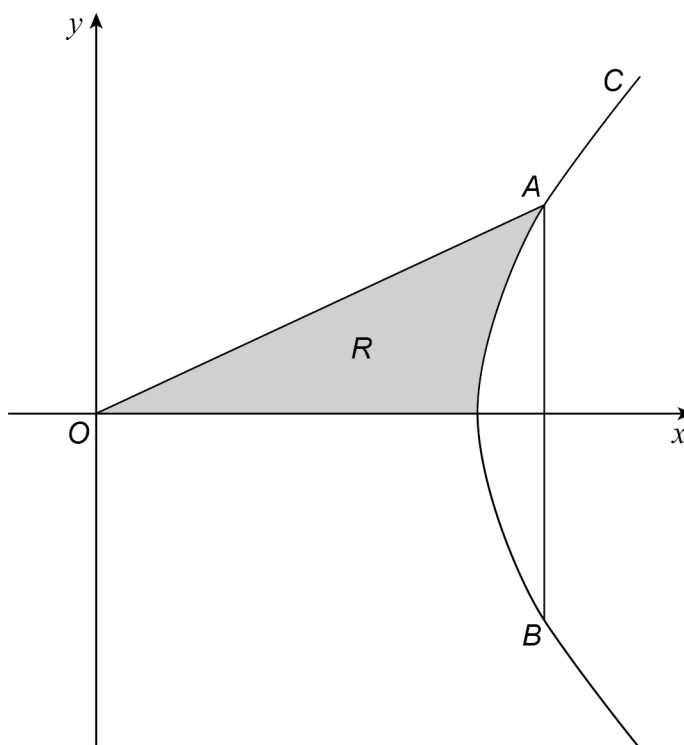
Turn over ►



13

The diagram shows the points A and B on the curve C and the lines OA and AB .

The point O is the origin and AB is perpendicular to the x -axis.



The Cartesian equation of curve C is $x^2 - y^2 = 8$ where $x \geq 2\sqrt{2}$.

The point A has Cartesian coordinates $(2\sqrt{3}, 2)$.

13 (a)

Taking the positive x -axis as the initial line and O as the pole, show that the polar equation of curve C is $r = \sqrt{8 \sec 2\theta}$

[3 marks]



Give your answer in the form $\ln(a + b\sqrt{3})$ where a and b are integers.

[illegible]

Give your answer in an exact form.

11

A curve C has equation

where k is a constant and $|k| < 1$

It is given that the line $y = 10$ is a tangent to the curve C

Show that the exact value of k can be expressed in the form $k = \frac{e+p}{e+q}$

where p and q are real numbers.

[10 marks]

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10



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